

Natural and Vaccine-Induced Immunity: Approximate Waning Rates for Major Vaccine-Preventable Infections

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Abstract

This note compares the duration of protection following natural infection with the duration of protection following vaccination for a selection of important infectious diseases, including COVID-19. The information is based primarily on official material from the U.S. Centers for Disease Control and Prevention (CDC) — chiefly the *Pink Book*, CDC’s reference text on vaccine-preventable diseases — together with World Health Organization (WHO) material where appropriate.

A numerical “mean duration of immunity” is not available for many diseases. Official sources frequently use expressions such as “lifelong,” “probably lifelong,” “at least 20 years,” or “waning over several years.” It would therefore be misleading to assign a spurious numerical mean to all diseases. The first part of the note reports the official qualitative or observational estimates, disease by disease. In the final table, these are converted – where, and only where, this is reasonably justified – into an indicative waning rate

$$\lambda = \frac{1}{T},$$

measured in day^{-1} , where T is a characteristic duration in days. Every table entry is coded using one of four symbols, defined immediately below the table: **0** (effectively lifelong, hence no finite waning rate), a **numerical value**, **N/D** (not determined precisely enough by the evidence), or **N/C** (not meaningfully reducible to a single characteristic rate).

1 Interpretation of the comparison

It is important to distinguish three related but different quantities:

1. protection against infection or reinfection;
2. protection against symptomatic disease; and
3. protection against severe disease, hospitalization, or death.

These quantities can have quite different durations. This is particularly important for influenza and COVID-19, for which immunity against infection can wane considerably faster than protection against severe disease.

Moreover, “waning” is not generally a simple exponential process. The quantity $\lambda = 1/T$ used in the table below is therefore a *characteristic inverse timescale*, not a directly measured biological decay constant. A genuine exponential model would require a well-defined survival or protection curve and would normally use a hazard or decay parameter estimated from longitudinal data. For several diseases discussed below, official sources decline to provide any such curve, and in those cases no number is invented.

2 Summary of official evidence

2.1 Measles

The CDC states that immunity following natural measles infection is lifelong; this observation dates at least to Peter Panum’s 1846 description of the disease (Ref. 7). For measles-containing vaccine, the CDC states that both serologic and epidemiologic evidence indicate that vaccine-induced immunity is long-term and probably lifelong in most persons. After two doses, more than 99% of persons develop serologic evidence of immunity, and persons who appear to lose antibody typically show a rapid (anamnestic) response on re-exposure, suggesting that protection persists even when circulating antibody is low (Ref. 7).

Thus, for both natural infection and vaccination, the appropriate description is essentially lifetime protection rather than a finite mean duration.

2.2 Mumps

Mumps is the clearest exception among the MMR-preventable diseases. Natural mumps infection is generally described as conferring lifelong immunity, but this characterization is somewhat softer than for measles or rubella: reinfections, though uncommon, are recognized to occur.

Vaccine-induced immunity is where mumps differs most sharply from measles and rubella. The CDC’s Pink Book chapter on mumps quantifies vaccine *effectiveness* (78% after one dose, 88% after two doses) but – unlike the measles, rubella, or polio chapters – does not offer any official statement of the *duration* of protection (Ref. 8). CDC’s current public-facing summary states only that “immunity against mumps may decrease over time” (Ref. 14). In the absence of an official duration, we draw on a widely cited peer-reviewed academic estimate: pooling six vaccine-effectiveness studies with a dynamic transmission model, Lewnard and Grad estimated that vaccine-derived protection against mumps wanes with a mean duration of 27 years (95% confidence interval, 16–51 years) after vaccination, and that this waning – rather than viral evolution – explains the resurgence of mumps among vaccinated young adults in the United States since 2006 (Ref. 16). This is an academic modeling estimate, not an official CDC or WHO figure, and it is flagged as such in the table below.

2.3 Rubella

The CDC reports that more than 90% of vaccinated persons have protection against clinical rubella for at least 15 years, and that follow-up studies indicate that one dose provides long-term, probably lifelong, protection. Although antibody titers wane in the years after vaccination, the CDC notes there is no evidence this leads to significant susceptibility to clinical rubella or congenital rubella syndrome (Ref. 11). Natural infection is generally regarded as producing long-lasting immunity.

Because the CDC’s own bottom-line characterization of the vaccine is “probably lifelong” – with the 15-year figure serving as supporting evidence for that conclusion rather than as the conclusion itself – rubella vaccine, like the measles vaccine, is treated below as falling in the “effectively lifelong” category rather than being assigned a finite rate. This differs from the earlier draft of this note, which had, inconsistently, converted the 15-year figure directly into a rate while treating comparably worded statements for other diseases as indicating zero waning; the revision here applies one rule uniformly, and it is discussed further below the table.

2.4 Varicella

Following vaccination, the CDC reports that immunity appears long-lasting and is probably permanent in the majority of recipients. More than 90% of vaccine responders maintain antibody for at least six years, with antibody also documented seven to ten years after vaccination in Japanese studies (Ref. 12). As with rubella, these interim data points support the CDC’s “probably permanent” conclusion rather than replacing it with a finite number.

Natural varicella infection generally produces lifetime immunity, although the precise duration is not represented by a single experimentally measured mean; second occurrences of chickenpox are uncommon in immunocompetent persons.

2.5 Hepatitis A

The CDC states plainly that the exact duration of protection following hepatitis A vaccination is unknown – a materially different, more open-ended statement than the “probably lifelong” language used for measles, rubella, varicella, or polio. Anti-HAV has persisted for at least 25 years in adults vaccinated as children under the older three-dose schedule used before 1999, and for at least 20 years in adults who received a two-dose schedule. Mathematical modeling and antibody-kinetic studies estimate that detectable antibody could persist for 40 years or longer (Ref. 2). Because the CDC explicitly withholds a “lifelong” characterization here, these figures are used below as the basis for a genuine (upper-bound) numerical rate.

Natural hepatitis A infection is generally regarded as producing durable, essentially lifelong immunity.

2.6 Hepatitis B

Hepatitis B vaccination produces long-lasting immunologic memory. The CDC reports that immune memory remains intact for more than 30 years following immunization, and that both adults and children with declining antibody levels remain protected against clinically significant infection because of a rapid anamnestic response on exposure. Booster doses are not recommended for immunocompetent persons (Ref. 3).

The interpretation of *natural* immunity is fundamentally more complicated, because infection can either resolve with immunity or become chronic, and these are qualitatively different outcomes rather than two points on the same waning curve. For this reason – discussed further below – natural hepatitis B immunity is not assigned a single characteristic rate at all; the duration of protective immunity after a resolved infection should in any case not be confused with the natural history of chronic hepatitis B infection itself.

2.7 Poliomyelitis

Natural poliovirus infection produces long-lasting immunity. The CDC describes immunity after a complete vaccination series – with either inactivated (IPV) or live oral (OPV) vaccine – as probably lifelong, while noting that the precise duration cannot be experimentally pinned down to a single mean value (Ref. 10).

2.8 Pertussis

Pertussis provides a particularly useful contrast. The CDC states explicitly that immunity following *Bordetella pertussis* infection is not permanent (Ref. 9). Published epidemiological estimates place

the duration of natural protection in the range of roughly four to twenty years, depending on the study and the definition of protection used.

Vaccine-induced immunity is shorter still. Protection following acellular pertussis vaccination (DTaP/Tdap) wanes appreciably over approximately four to five years, which is a principal reason booster doses are recommended (Ref. 9).

2.9 Influenza

Influenza differs fundamentally from the diseases above because immunity is shaped not only by waning immune responses but also by the antigenic evolution (“drift”) of the circulating virus. The CDC states explicitly, as an official practical conclusion, that “for practical purposes, the duration of immunity following influenza vaccination is less than one year because of vaccine-induced antibody waning and antigenic drift of circulating influenza viruses” (Ref. 6). Because this is an explicit, official, single-number statement, it is used below as a genuine (if composite) rate for the vaccine.

No comparable official statement exists for the duration of *natural* post-infection immunity to influenza. This is not merely an oversight: a single number would conflate true immunological waning (which, against an unchanged strain, may be prolonged) with antigenic drift of the virus itself (which can defeat even fully intact immunity within a season). In the absence of an official figure isolating the two effects, natural influenza immunity is left undetermined below rather than assumed equal to the vaccine figure, as the earlier draft of this note had done.

2.10 Human papillomavirus (HPV)

Natural HPV infection does not reliably produce durable protective immunity against subsequent infection; reinfection and infection with additional HPV types can occur. This is a qualitatively different situation from “fast waning” – for many infections, robust cross-protective immunity is never clearly established in the first place, so there is no single protection curve to characterize.

HPV vaccination, by contrast, produces strong and durable antibody responses. The CDC’s Pink Book states that ongoing monitoring has demonstrated vaccine effectiveness above 90%, with no waning of immunity through at least 10 to 12 years of follow-up (Ref. 4), consistent with CDC’s public-facing statement that protection lasts more than 10 years without becoming less effective and that no data indicate protection lessens over time (Ref. 5). As with hepatitis A, this is phrased as an absence of observed waning within a follow-up window, not as an affirmative “lifelong” finding, so it is used below as the basis for a genuine (upper-bound) numerical rate rather than being coded as zero.

2.11 COVID-19

COVID-19 requires particular caution. Natural infection produces immune responses in most individuals, but protection against reinfection changes with time and with the emergence of antigenically different variants. The WHO’s 2021 scientific brief on natural immunity reported that, in the evidence available at the time, most infected people had robust protective immune responses for at least 6–8 months, while explicitly cautioning that this finding “should not be interpreted as a fixed biological expiration date” and flagging that variants of concern could alter the picture (Ref. 15).

Vaccine-induced protection shows a similar pattern but is explicitly bifurcated by endpoint in official communications: protection against infection wanes over a period of months, while protection against severe disease, hospitalization, and death generally persists substantially longer. Updated

vaccines are recommended precisely because both immune waning and antigenic evolution alter protection over time.

Unlike influenza, no official source commits to a single practical duration figure for COVID-19 that spans both endpoints and both exposure routes. For this reason, neither natural nor vaccine-induced COVID-19 immunity is reduced to a single rate in the table below; both are coded as not meaningfully comparable via one characteristic waning rate, which is a stronger conclusion than simply lacking a number, and is consistent with how official sources themselves discuss COVID-19 immunity.

2.12 A note on the stability of “official” sources in 2025–2026

Between 2025 and 2026, U.S. federal vaccine recommendations went through an unusually turbulent period: the Secretary of Health and Human Services replaced the full membership of the Advisory Committee on Immunization Practices (ACIP) in 2025, ACIP subsequently revised several recommendations (including moving the birth-dose hepatitis B recommendation from universal to shared clinical decision-making, discontinuing preference for the combination MMRV vaccine, and recommending a single-dose HPV schedule), and a federal court stayed both the resulting 2026 childhood immunization schedule and the disputed ACIP appointments pending litigation (Ref. 17). These disputes concern vaccination *schedules and policy*, not the underlying serologic and epidemiologic evidence on duration of protection summarized above, which predates this period and is unaffected by it. Readers should nonetheless be aware that “the official CDC position” has been an unusually moving target in this specific window, and should check current primary sources for anything schedule-related.

3 Why inverse duration is used

Suppose, purely as a phenomenological approximation, that a protection timescale is represented by T days. We define the corresponding characteristic waning rate by

$$\lambda = \frac{1}{T} \text{ day}^{-1}.$$

This definition is useful for comparing orders of magnitude. For example,

T	$1/T \text{ (day}^{-1}\text{)}$
1 year (365 days)	2.74×10^{-3}
5 years	5.48×10^{-4}
10 years	2.74×10^{-4}
20 years	1.37×10^{-4}
27 years	1.01×10^{-4}
40 years	6.85×10^{-5}

(All year-to-day conversions in this note use 365 days per year, for consistency; this introduces no meaningful error at the precision used here.) These numbers should be read as inverse characteristic durations, not as experimentally established exponential decay constants.

For lifelong immunity, it is more scientifically honest to write “the waning rate is effectively zero on the finite epidemiological timescales considered” than to assign an arbitrary human life expectancy as T . Accordingly, the numerical table below uses the symbol 0 whenever the official source’s own bottom-line characterization is “lifelong” or “probably lifelong,” even if a finite interim duration is cited elsewhere in the source as supporting evidence for that conclusion.

4 Numerical comparison

The table below is deliberately conservative and follows one rule throughout: a number is entered only when the cited source’s own substantive conclusion is a finite, bounded duration – not when a finite duration merely appears as an intermediate data point supporting a “lifelong” conclusion, and not when the underlying process cannot be sensibly reduced to a single rate at all. Every cell in the last three columns is coded using exactly one of four symbols, defined immediately below the table. Diseases are grouped by the value of the ratio in the final column, in the order 0, then a nonzero number, then N/C, then N/D.

Table 1: Indicative waning rates, writing $\theta \equiv \lambda_{\text{natural}} = 1/T_{\text{natural}}$ for the characteristic natural-immunity waning rate and $\psi \equiv \lambda_{\text{vaccine}} = 1/T_{\text{vaccine}}$ for the characteristic vaccine-immunity waning rate (both in day^{-1}). Diseases are grouped by the value of θ/ψ , coded as described below the table.

Disease	Natural immunity: official characterization	Vaccine immunity: official characterization	Natural θ	Vaccine ψ	θ/ψ
Group 1: $\theta/\psi = 0$ (natural immunity effectively non-waning; vaccine wanes at a finite rate)					
Mumps	Generally lifelong, but reinfection documented more often than for measles/rubella	No official duration; academic model: mean 27 yr (95% CI 16–51 yr)	0	1.0×10^{-4} ($0.5\text{--}1.7 \times 10^{-4}$)	0
Hepatitis A	Essentially lifelong	CDC: exact duration unknown; documented $\geq 20\text{--}25$ yr; modeled ≥ 40 yr	0	$\leq 1.37 \times 10^{-4}$ (as low as 6.85×10^{-5} under modeling)	0
Group 2: θ/ψ a nonzero number (both mechanisms wane at a finite rate)					
Pertussis	CDC: not permanent; published epidemiological estimates $\approx 4\text{--}20$ yr	Wanes appreciably over $\approx 4\text{--}5$ yr (acellular vaccine)	$1.37 \times 10^{-4}\text{--}6.85 \times 10^{-4}$	$5.48 \times 10^{-4}\text{--}6.85 \times 10^{-4}$	$\approx 0.2\text{--}1.25$
Group 3: $\theta/\psi = \text{N/C}$ (not meaningfully comparable via one characteristic rate)					
<i>(continued on next page)</i>					

Disease	Natural immunity: official characterization	Vaccine immunity: official characterization	Natural θ	Vaccine ψ	θ/ψ
Measles	Lifelong (Panum, 1846)	Long-term, probably lifelong; >99% seroconvert after 2 doses	0	0	N/C
Rubella	Long-lasting / essentially lifelong	≥ 15 yr documented; officially “probably lifelong”	0	0	N/C
Varicella	Long-lasting; second episodes uncommon	Antibody documented $\geq 6-10$ yr; officially “probably permanent”	0	0	N/C
Poliomyelitis	Long-lasting	IPV and OPV both officially “probably lifelong”	0	0	N/C
Hepatitis B	Heterogeneous: resolves with immunity, or becomes chronic; not one process	Immune memory documented ≥ 30 yr; boosters not advised in immunocompetent adults	N/C	$\leq 9.13 \times 10^{-5}$	N/C
HPV	No reliable durable natural protection; not a simple waning process	CDC: >90% effective, no waning observed through 10–12 yr	N/C	$\leq 2.74 \times 10^{-4}$	N/C

(continued on next page)

Disease	Natural immunity: official characterization	Vaccine immunity: official characterization	Natural θ	Vaccine ψ	θ/ψ
COVID-19	WHO (2021): most had robust responses $\geq 6-8$ mo against reinfection; explicitly not a fixed expiry; variant-dependent	Protection vs. infection wanes over months; vs. severe disease persists longer; no single official figure	N/C	N/C	N/C
Group 4: $\theta/\psi = \text{N/D}$ (not determined precisely enough by official evidence)					
Influenza	No official duration; confounded with antigenic drift	CDC: practically <1 yr, from waning <i>and</i> antigenic drift	N/D	$\gtrsim 2.74 \times 10^{-3}$	N/D

Key to the table

0 Effectively lifelong protection, hence a zero waning rate on the epidemiological timescales considered here. Used whenever the cited source’s own substantive conclusion is “lifelong” or “probably lifelong,” even if a shorter documented interval is cited elsewhere in the same source as supporting evidence.

numerical value

A defensible finite characteristic rate. Used only when the cited source’s own conclusion is itself a bounded, finite duration (an observed floor, a modeled estimate, or an explicit practical figure) rather than an affirmation of durability. Where the underlying duration is itself a range or a confidence interval, the corresponding range of θ or ψ is given.

N/D Not determined sufficiently well by the official evidence. Used when a finite characteristic duration plausibly exists in principle but no official source states or bounds it (natural influenza immunity is the clearest case here).

N/C Not meaningfully comparable using one characteristic waning rate. Used when the underlying phenomenon cannot be sensibly reduced to a single decay parameter at all – for example, because natural infection outcomes are qualitatively heterogeneous (hepatitis B), because natural infection does not reliably establish protective immunity in the first place (HPV), because two protection endpoints diverge too far to summarize in one number (COVID-19), or because a ratio would otherwise involve dividing by, or into, an undefined or ill-posed quantity.

A methodological note: several of these classifications required a judgment call, and a reasonable person could draw some of these lines differently – most notably for rubella and varicella vaccine,

discussed in Sections 2.3 and 2.4 above, where an interim documented duration exists alongside an official “probably lifelong” conclusion. The rule applied throughout this revision is to code according to the source’s own substantive conclusion, not the most conservative number appearing anywhere in its supporting text; this is a change from the earlier version of this note, which applied that rule inconsistently across diseases.

The ratio column, θ/ψ , follows directly from the coding of its numerator θ and denominator ψ : a ratio of exactly 0 is entered only when θ is coded 0 and ψ is a genuine finite number (giving a mathematically exact zero, not merely an approximation); the ratio is coded N/C whenever *either* θ or ψ is itself 0 paired with 0, or is coded N/C; and it is coded N/D when either component is coded N/D. Note that this differs from, and corrects, the earlier version of this table, which entered “ ∞ ” for several diseases where both rates were effectively zero – a case where θ/ψ is actually indeterminate (0/0), not infinite.

Important qualification. The COVID-19 and influenza entries require special care. Even where a number is given for influenza vaccine (the one case with an explicit official practical figure), it reflects a mixture of true immunological waning and antigenic drift of the circulating virus, not a pure decay constant. The COVID-19 figures discussed in the text above are deliberately *not* carried into the table as numbers, precisely because official sources themselves decline to summarize COVID-19 immunity with one figure; none of this should be read as implying that protection disappears abruptly at any of the quoted intervals.

5 Official references

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5. CDC, *HPV Vaccine Safety and Effectiveness Data*. States that protection lasts more than 10 years without becoming less effective and that no data indicate waning over time. <https://www.cdc.gov/hpv/hcp/vaccination-considerations/safety-and-effectiveness-data.html>

6. CDC, *Pink Book, Chapter 12: Influenza*. States that, for practical purposes, the duration of immunity following influenza vaccination is less than one year, due to a combination of antibody waning and antigenic drift. <https://www.cdc.gov/pinkbook/hcp/table-of-contents/chapter-12-influenza.html>
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13. CDC, *Pink Book, Site Index*. Full table of contents for the Pink Book's chapters on individual vaccine-preventable diseases. <https://www.cdc.gov/pinkbook/site.html>
14. CDC, *Measles Vaccination*. States that MMR/MMRV vaccines usually protect against measles and rubella for life, while immunity against mumps may decrease over time. <https://www.cdc.gov/measles/vaccines/index.html>
15. WHO, *COVID-19 Natural Immunity: Scientific Brief*, 10 May 2021. Reports that most infected persons had robust protective immune responses for at least 6–8 months in the evidence available at that time, while cautioning against interpreting this as a fixed expiration date. <https://iris.who.int/handle/10665/341241>
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6 Concluding observation

The principal lesson from the comparison is that the ratio θ/ψ (writing $\theta \equiv \lambda_{\text{natural}}$ and $\psi \equiv \lambda_{\text{vaccine}}$, as in Table 1), where it can be formed at all, is highly disease-dependent. There is no universal biological principle according to which natural immunity must wane more slowly than vaccine-induced immunity, nor the converse.

For measles, rubella, varicella, and polio, both natural infection and vaccination are officially characterized as producing extremely durable, essentially non-waning immunity, and the ratio between them is not meaningfully defined. For mumps and hepatitis A, natural infection is essentially non-waning while the vaccine wanes on a finite (multi-decade) timescale, giving a ratio of exactly zero. For pertussis, both mechanisms wane on a finite timescale, with natural immunity lasting somewhat longer on average than protection from the current acellular vaccines, though neither is permanent. For HPV, natural infection is a poor and inconsistent source of durable protection, whereas vaccination produces very durable protection – the opposite pattern from pertussis, and one where the natural-immunity side of the ratio is not well-posed at all. For hepatitis B, the very idea of a single natural-immunity waning rate breaks down because infection outcomes are qualitatively heterogeneous. For influenza and COVID-19, waning immunity and antigenic evolution are entangled to the point that official sources themselves decline to offer a single duration, and this note follows their lead rather than manufacturing false precision.

Consequently, the inverse-duration table is best regarded as a compact phenomenological summary useful for modeling in the cases where the underlying evidence supports it, and as an explicit record of where it does not, rather than as a table of fundamental immunological decay constants.